

3.7 Saving and Loading Files

Making data outlive the program

3.7 / THE PROBLEM

EVERYTHING SO FAR DISAPPEARS

Everything we've built lives only in the program's **memory** - it disappears the moment the program ends.

File I/O is how a program remembers something between separate runs.

WRITING A FILE

`std::ofstream` ("output file stream") opens a file for writing.



```
#include <fstream>

std::ofstream outputFile("friends.txt");

for (const std::string& friendName : friends) {
    outputFile << friendName << "\n";
}

outputFile.close();
```

Write to it with `<<`, just like `std::cout`.

READING A FILE BACK

`std::ifstream` ("input file stream") opens a file for reading.



```
std::ifstream inputFile("friends.txt");
std::vector<std::string> friendsFromDisk;

std::string line;
while (std::getline(inputFile, line)) {
    friendsFromDisk.push_back(line);
}

inputFile.close();
```

Same `std::getline` already used for keyboard input.

IT'S ALL JUST A STREAM

A file and the keyboard are both just a **stream of text** as far as C++ is concerned.

DIRECTION	TOOL
keyboard in	<code>std::cin</code>
console out	<code>std::cout</code> / <code>std::print</code>
file in	<code>std::ifstream</code>
file out	<code>std::ofstream</code>

PROVING IT WORKED



```
std::println("\nReloaded from {}:", fileName);  
for (const std::string& friendName : friendsFromDisk) {  
    std::println(" - {}", friendName);  
}
```

Written this run, read back this same run - next lecture makes it actually persist **across** runs.

[SCREENSHOT]

WATCH THE FILE APPEAR

Step over the `saveFriendsToFile`-equivalent write, then open `friends.txt` in your IDE's file browser - it now exists, with the friend names written to it.

[SCREENSHOT]

Next: 3.8

Remembering Things